

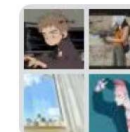
Instructor and TAs



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- 15:00 – 16:00 Monday
- Please send me an email before coming



该二维码7天内(3月29日前)有效, 重新进入将更新



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Lecture 05: Text Classification

Zhuyu Lin
Spring, 2026

Outline



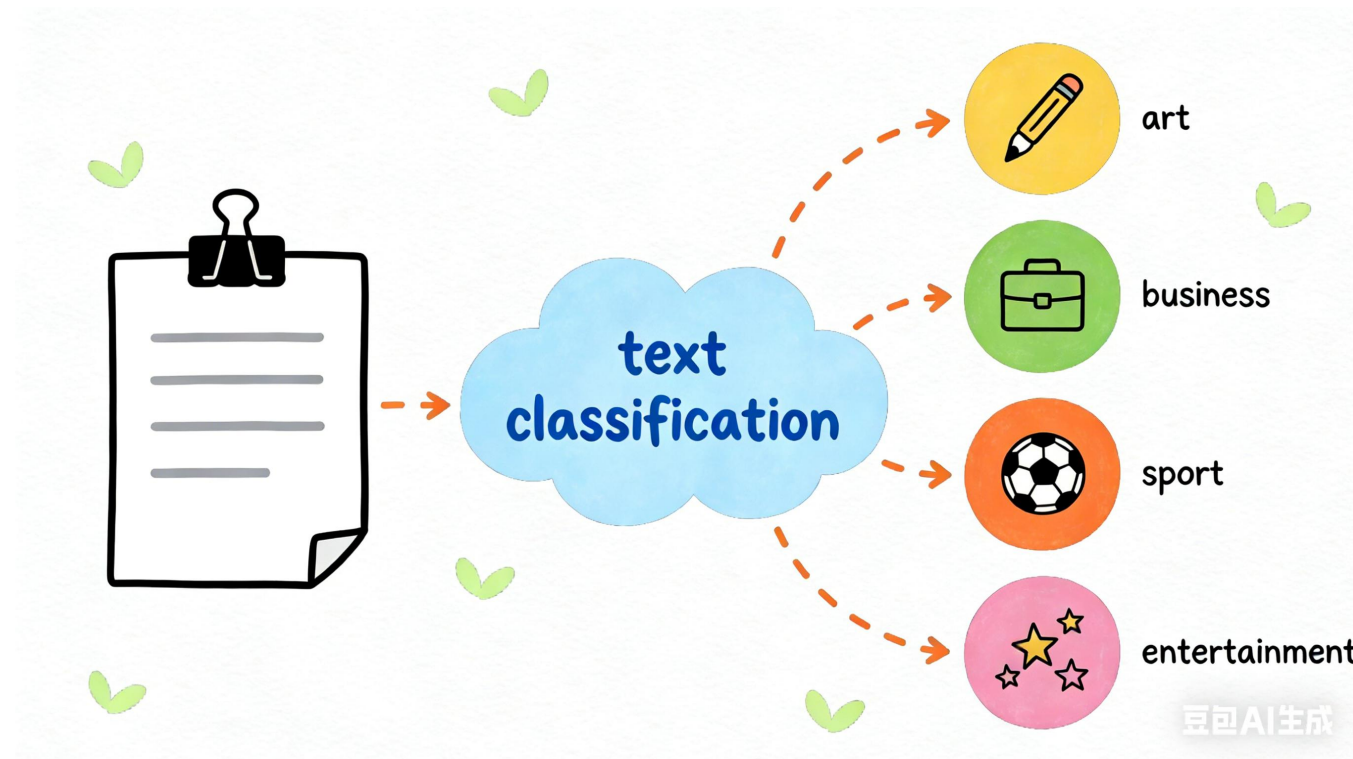
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- Introduction
- Methods
- Deep Learning for Text Classification
- Evaluation Metrics

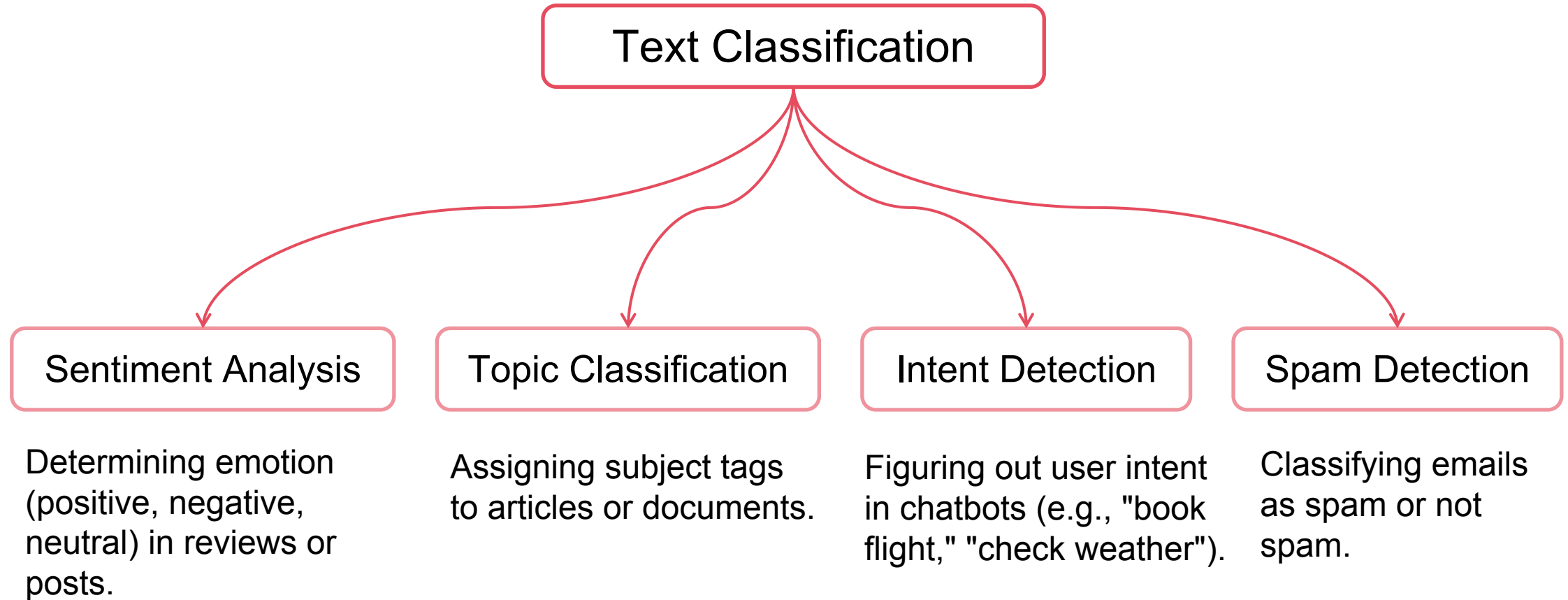
Text Classification



- Text classification is the process of assigning a predefined label or class to text.
- It is one of the most fundamental and widely used NLP tasks.



Text Classification



Text Classification: Sentiment Analysis



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My experience
so far has been
fantastic!

POSITIVE



The product is
okay I guess.

NEUTRAL



Your support
team is
useless.

NEGATIVE

Text Classification: Sentiment Analysis



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■ Practical applications

□ Help make better business decisions

- Group reviews by sentiment to analyze the text and make strategic decisions.

□ Personalized recommendations

- Suggest content based on emotional tone preferences.

■ Datasets

Dataset	Language	Number of labels	Size
IMDB Movie Review	English	2	50K
Amazon Reviews 2023	English	5	571.54M
ChnSentiCorp	Chinese	2	12K

Text Classification: Sentiment Analysis



■ Datasets

□ ChnSentiCorp examples

text string · lengths 	label class label 
选择珠江花园的原因就是方便，有电动扶梯直接到达海边，周围餐馆、食廊、商场、超市、摊位一应俱全。酒店装修一般，但还算整洁。泳池在大堂的屋顶，因此很小，不过女儿倒是喜欢。包的早餐是西式的，还算...	1 positive
15.4寸笔记本的键盘确实爽，基本跟台式机差不多了，蛮喜欢数字小键盘，输数字特方便，样子也很美观，做工也相当不错	1 positive
房间太小。其他的都一般。。。。。。。。	0 negative
1.接电源没有几分钟，电源适配器热的不行。2.摄像头用不起来。3.机盖的钢琴漆，手不能摸，一摸一个印。4.硬盘分区不好办。	0 negative

Text Classification: Topic Classification



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发现

分类 ^

电台

专辑/声音/主播



上传



创作中心



有声出版



开通会员



客户端



音乐

有声书

娱乐

外语

儿童

商业财经

历史

相声评书

个人成长

广播剧

有声图书

人文国学

热点

生活

悬疑

健康

汽车



儿童



历史



热点

■ Practical applications

□ Content indexing & organization

- Documents, web pages, videos, or products are automatically tagged with topics.
- Enables faster retrieval: Instead of scanning full text, the system matches based on precomputed topic labels.
- Supports faceted search: Users can filter results by topic (e.g., “Show only news in Politics”).

□ Query understanding in search systems

- Classifying a user’s search query into predefined topics (e.g., “sports,” “technology,” “health”) helps infer user intent.

□ Personalized recommendations

- User profiles are enriched with topic preferences derived from past interactions (clicks, views, purchases).
- Example: A user who frequently reads articles about “AI” and “machine learning” gets recommended new content in the Technology → AI topic.

Text Classification: Topic Classification



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■ Datasets

Dataset	Language	Number of labels	Size
AG News	English	4	128K
DBpedia_14	English	14	630K
THUCNews	Chinese	14	740K
IFlyTek	Chinese	119	17.3K

Text Classification: Topic Classification

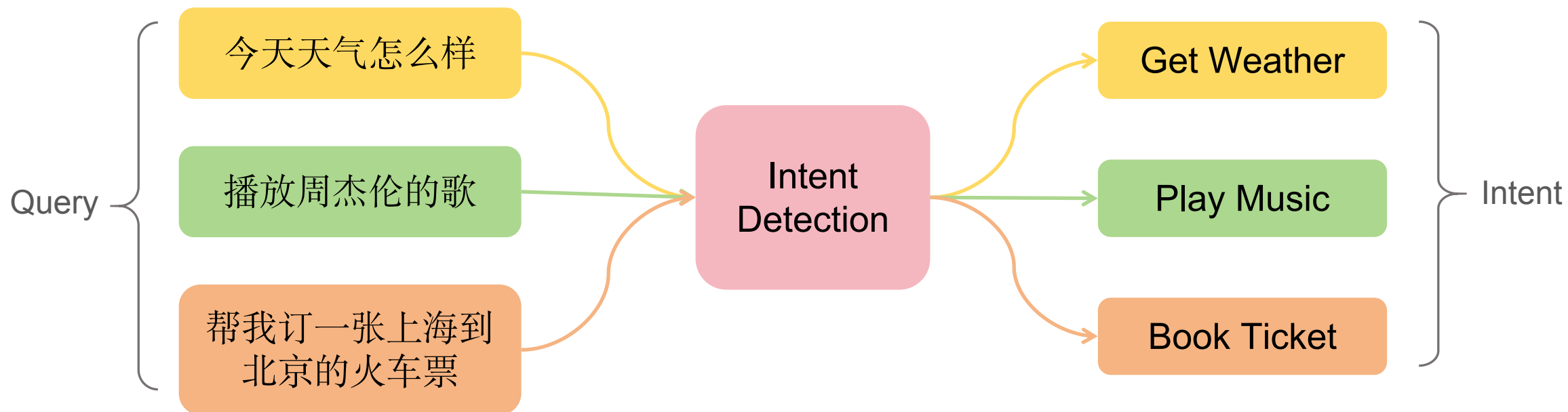


■ Datasets

□ AG News examples

text string · lengths	label class label
	
No Gold for Phelps as Relay Team Falters South Africa won the gold medal Sunday in the men's 400-meter freestyle relay with a world-record time of 3 minutes 13.17 seconds.	1 Sports
I Confess. I'm a Software Pirate. \\I'm guilty. I'm a Software Pirate! Not just one or two copies of DOS but\\probably hundreds of thousands of dollars worth of...	3 Sci/Tech
Venezuela Oil Official Predicts Stability (AP) AP - Venezuela's oil industry will enter a new period of stability and growth after Sunday's referendum on Hugo Chavez's presidency...	0 World
Funds: Are Bond Funds Hazardous? (Clint Willis is a freelance writer who covers mutual funds for Reuters. Any opinions in the column are solely those of Mr. Willis.)	2 Business

Text Classification: Intent Detection



■ Practical applications

□ Voice Assistants & Smart Devices:

- Voice assistant rely heavily on intent detection. The detected intent is what allows the system to execute the correct action.
- Examples: 小爱同学、Siri、In-vehicle voice assistant

□ Intelligent Customer Service:

- Systems analyze a user's query (e.g., "I need to reset my password", "What's my balance?") and identify the underlying goal (reset_password, check_balance).

Text Classification: Intent Detection



■ Datasets

Dataset	Description	Language	Number of labels	Size
ATIS (Airline Travel Information System)	A collection of utterances to a flight booking system	English	20+	5.87K
BANKING77	Online banking queries annotated with their corresponding intents.	English	77	13.1K
CLINC150	150 intent classes over 10 domains	English	150	23.7K
HWU64	Popular personal assistant queries	English	64	10.2K

Text Classification: Intent Detection

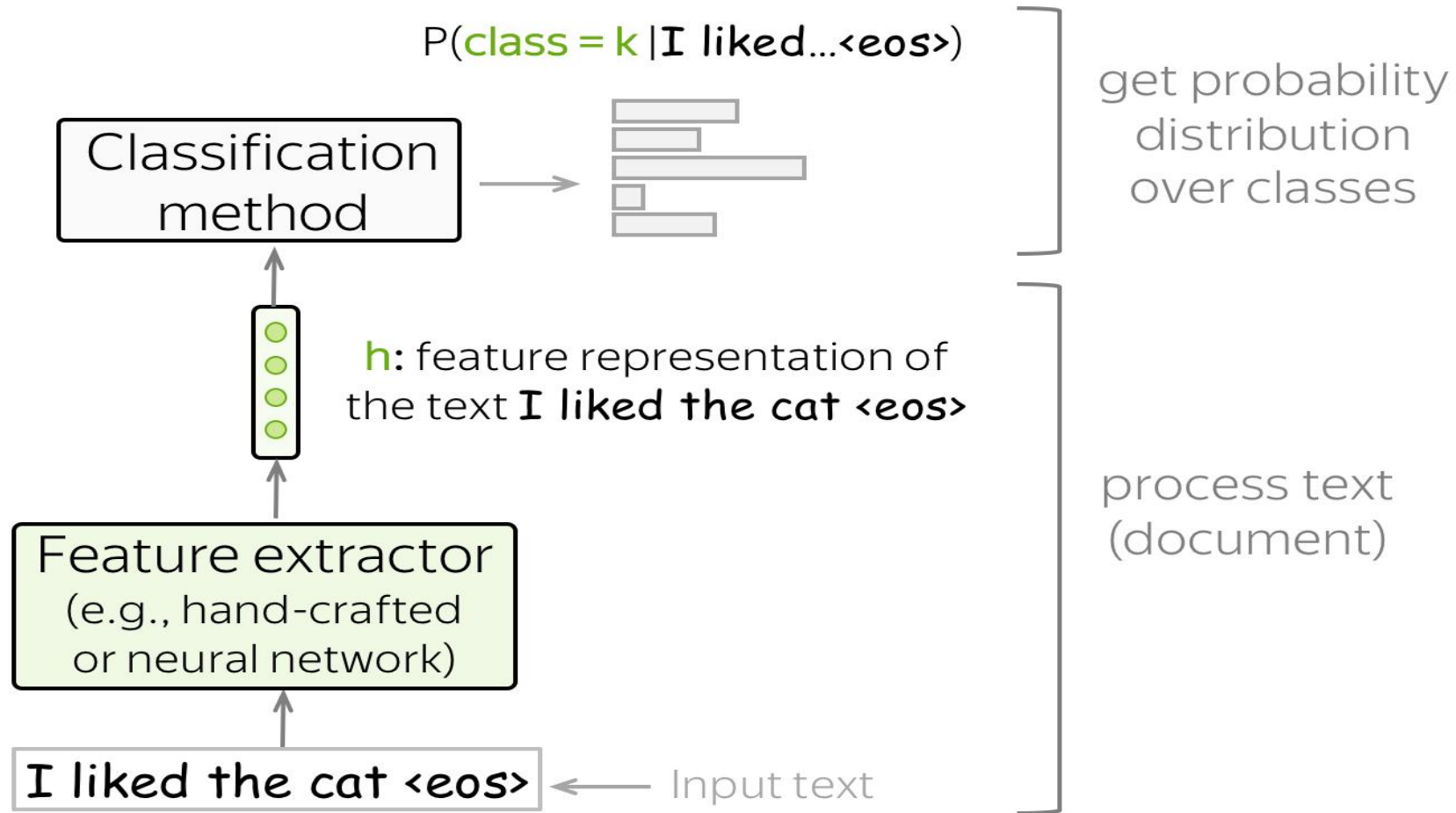


■ Datasets

□ CLINC150 examples

text string · <i>lengths</i>	domain class label	intent class label
		
what ingredients are in antipasto salad	3 kitchen_and_dining	34 kitchen_and_dining:ingredients_list
is there enough money in my bank of hawaii for vacation	1 banking	12 banking:balance
what's the nutrient info for a donut	3 kitchen_and_dining	36 kitchen_and_dining:nutrition_info
tell me any events scheduled on my calendar for april 15	4 home	55 home:calendar
check and confirm reservations at ruth chris for carol lee	3 kitchen_and_dining	32 kitchen_and_dining:confirm_reservation
what should i do if my card is lost	2 credit_cards	20 credit_cards:report_lost_card
what's the nutrient make up of a plate of spaghetti	3 kitchen_and_dining	36 kitchen_and_dining:nutrition_info
i need to let my bank know i am visiting asia soon	6 travel	77 travel:travel_notification

Text Classification



https://lena-voita.github.io/nlp_course/text_classification.html

Text Classification



- We assume that we have a collection of documents with ground-truth labels.
- Input of a classifier: a document $\mathbf{x} = (x_1, \dots, x_n)$ with tokens $(x_1, \dots, x_n)$
- Output of a classifier: a label $y \in Y = \{y_1, \dots, y_K\}$
- A classifier estimates probability distribution $p(y|\mathbf{x})$
- Pick a class with the highest conditional probability:

$$y = \operatorname{argmax}_k p(y = k|\mathbf{x})$$

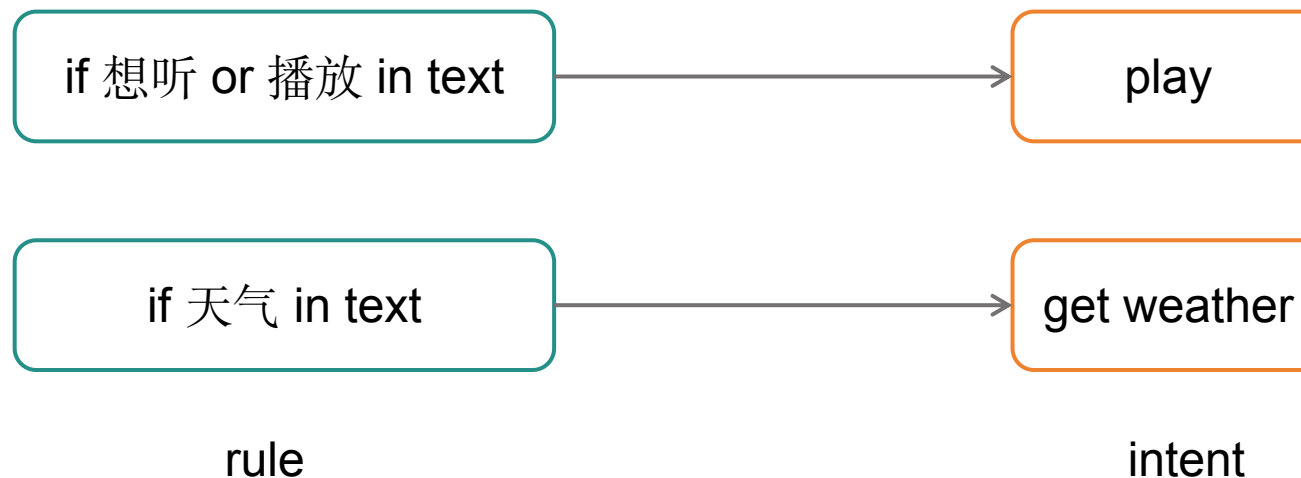
Text Classification: Methods



- Rule-Based Systems
- Traditional Machine Learning
- Deep Learning(CNN/RNN)
- Deep Learning(Transformer-Based Models)
- LLMs(Few-Shot / Zero-Shot Learning)

■ Rule-Based Systems

- Use manually crafted linguistic rules (e.g., keyword matching, regular expressions, pattern-based rules).
- Pros: easy to implement for simple tasks, no training data needed.
- Cons: Not scalable, brittle (can't handle unseen cases), requires domain expertise to create and maintain.



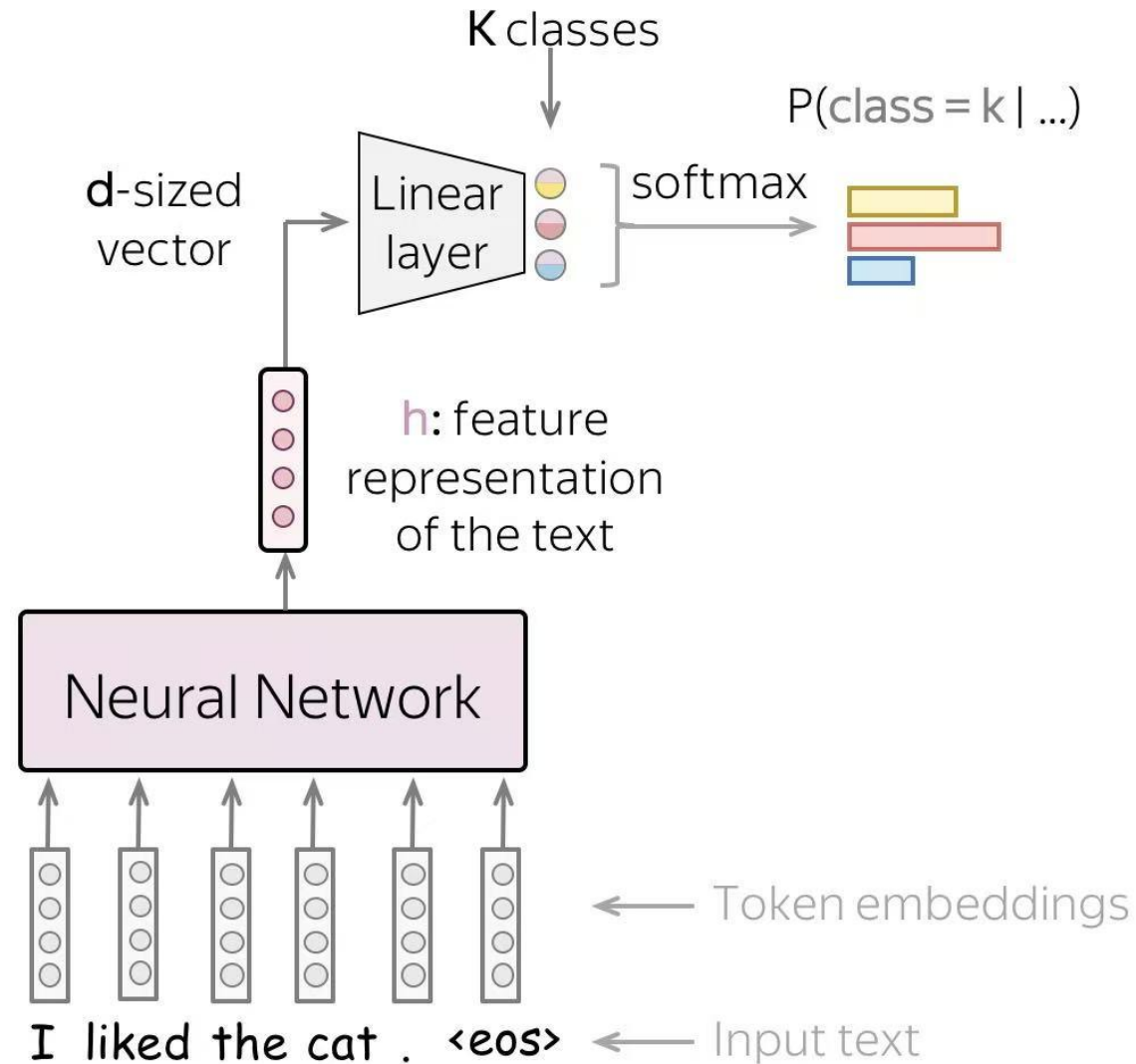
■ Traditional Machine Learning

- Requires feature extraction from text (converting text to numbers) before applying standard ML algorithms.
- Common Algorithms: Naive Bayes, Logistic Regression, SVM, Decision Trees / Random Forests.
- Pros: Often lighter and faster than deep learning. Can work well with smaller datasets. Models are generally interpretable.
- Cons: Performance heavily depends on feature engineering. Cannot capture complex semantic meaning or long-range context.

Text Classification: Methods



■ Deep Learning



■ Deep Learning(RNN/CNN)

□ Before Transformers, state-of-the-art for most complex NLP tasks.

□ Pros:

- No need for manual feature engineering (like TF-IDF, handcrafted n-grams).
- The network learns meaningful hierarchical representations directly from raw text or word embeddings.
- Significantly outperforming traditional ML on large datasets.

□ Cons:

- Much heavier than traditional ML models like Naive Bayes or SVM.
- Training from scratch, to generalize well and avoid overfitting, they typically need large amounts of labeled training data.
- Less interpretable ("black-box").

■ Deep Learning(Transformer-Based Models)

- State-of-the-art performance, widely used
- **Pre-trained** on massive text corpora. Then **fine-tuned** on specific classification tasks.
- Models: BERT, RoBERTa, DistilBERT, and the GPT family.
- Pros:
 - Achieve superior accuracy, understand nuance, context, and polysemy (words with multiple meanings).
 - For a specific task (like classification), you only need to fine-tune the pre-trained model on a relatively small labeled dataset, achieving high accuracy quickly.
- Cons:
 - Computationally expensive
 - Latency increases quadratically with token count → long documents become very slow.

■ LLMs (Few-Shot / Zero-Shot Learning)

□ LLMs can perform classification based on instructions and a few examples provided in a **prompt**, **without explicit fine-tuning**.

□ Pros:

- Very little or no labeled training data.
- No need for thousands of labeled examples.
- Test classification ideas in minutes without data collection/annotation.

□ Cons:

- Inference is expensive
- High Latency.

请对以下新闻进行分类，类别选项有：时政、体育、科技、财经、社会、娱乐、军事、其他。只给出分类结果，不要输出其他内容。

新闻：近期贵金属原材料价格大幅攀升，中国金币集团发布风险提示，提醒广大贵金属纪念币集藏爱好者，理性参与投资，注意管控风险。

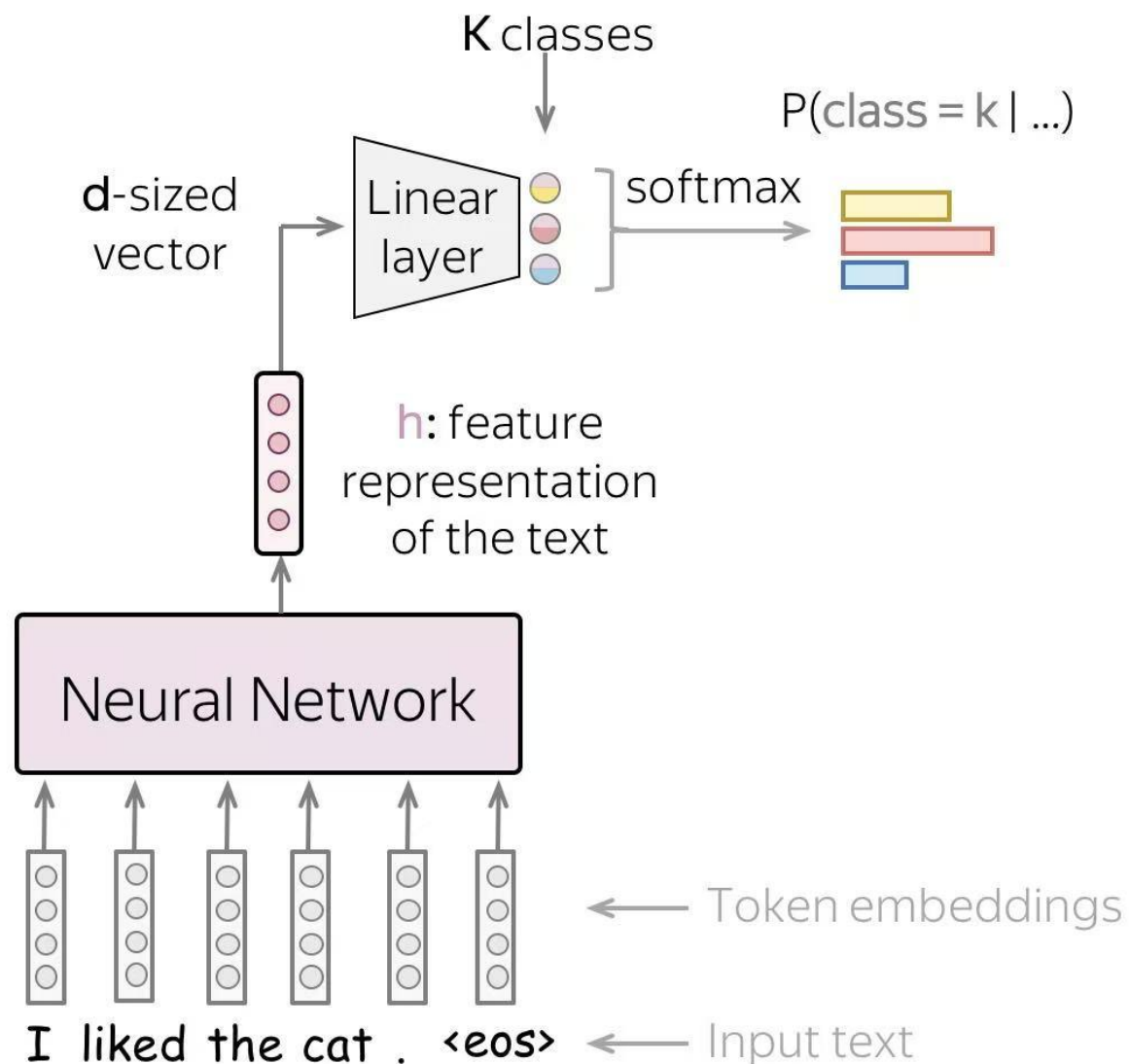


☒ 已思考 (用时 12 秒) >

财经

本回答由 AI 生成，内容仅供参考，请仔细甄别。

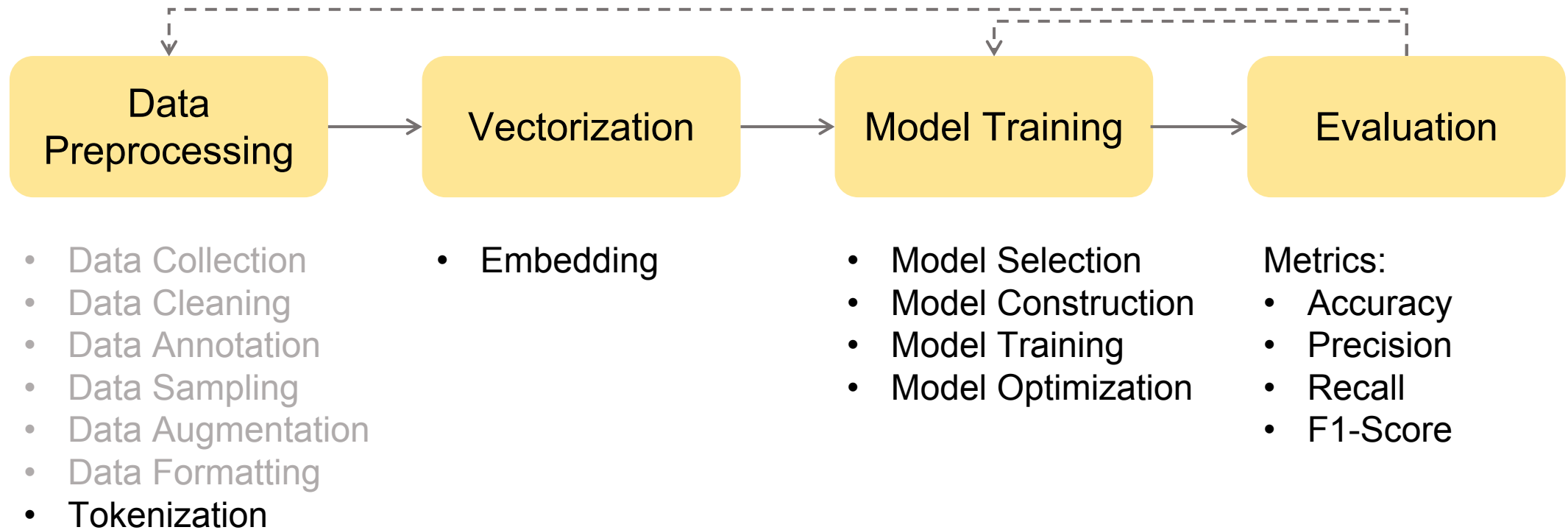
Deep Learning for Text Classification



Deep Learning for Text Classification



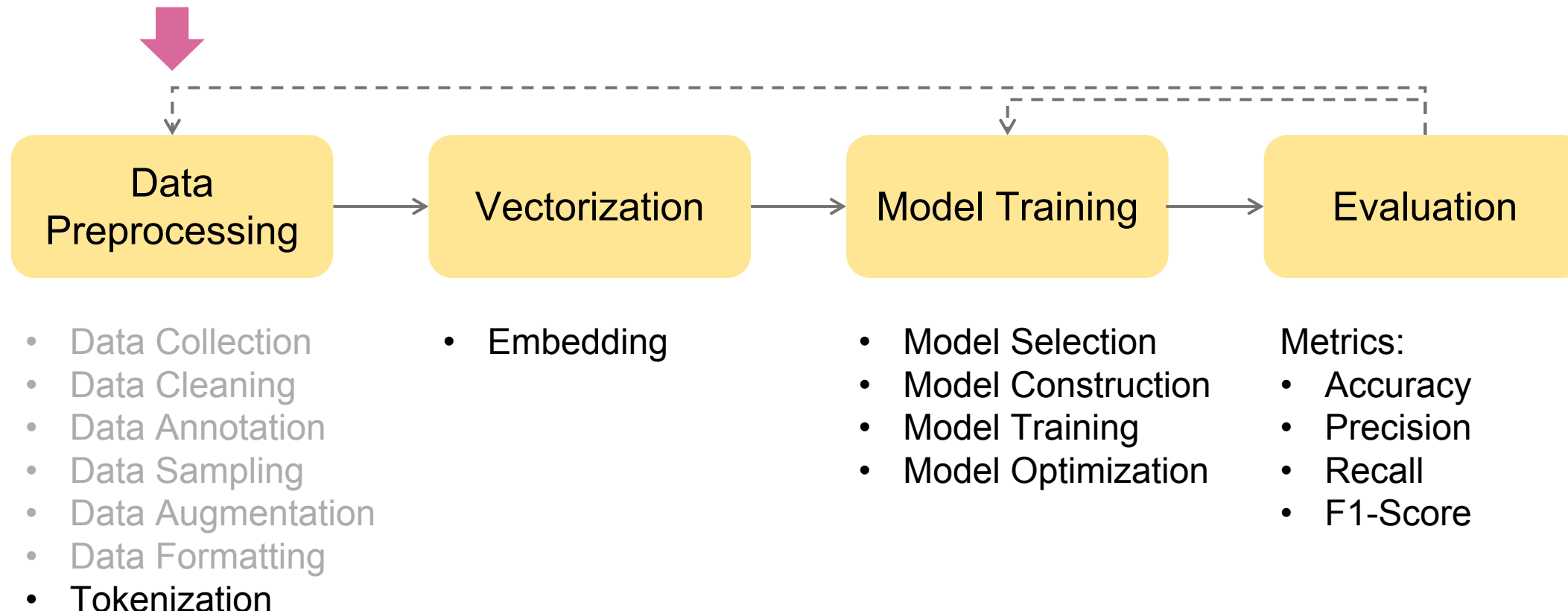
■ Pipeline



Deep Learning for Text Classification



■ Pipeline



■ Tokenization

1. Split text into **tokens** (words, subwords, or characters)

- English: whitespace + punctuation handling
- Chinese: use segmentation tools (e.g., Jieba, spaCy with zh model)
- Modern models often use subword tokenization (e.g., BPE, WordPiece, SentencePiece)

2. Build a **vocabulary** based on the training corpus

- Vocabulary: the set of unique tokens present in a training corpus, forming a fixed dictionary used to convert text into numerical representations for models.
- Includes special tokens such as [UNK] (unknown), [PAD] (padding), [CLS] (start), or [SEP] (separator).

3. Map **tokens to IDs** for model input

Deep Learning for Text Classification



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■ Tokenization

1. Split text into tokens (words, subwords, or characters):

input text

剧情紧凑，演员演技在线，强烈推荐!

word-based

剧情 / 紧凑 / ， / 演员 / 演技 / 在线 / ， / 强烈 / 推荐 / !

character-based

剧 / 情 / 紧 / 凑 / ， / 演 / 员 / 演 / 技 / 在 / 线 / ， / 强 / 烈 / 推 / 荐 / !

■ Tokenization

1. Split text into tokens (words, subwords, or characters):

- Subword tokenization algorithms rely on the principle that frequently used words should not be split into smaller subwords, but rare words should be decomposed into meaningful subwords.

input text	Let's do tokenization!																			
word-based	Let				's			do			tokenization					!				
character-based	L	e	t	'	s	d	o	t	o	k	e	n	i	z	a	t	i	o	n	!
subword-based	Let's </w>					do</w>				token			ization</w>					!</w>		

■Tokenization

2. Build a **vocabulary** based on the training corpus

- Vocabulary: the set of unique tokens present in a training corpus, forming a fixed dictionary used to convert text into numerical representations for models.

text
剧情紧凑，演员演技在线，强烈推荐!
剧情老套，台词尴尬，毫无新意。
虽然是小成本制作，但情感真挚，细节打动人心。
节奏慢得像蜗牛，两个小时都在打瞌睡。
.....

tokenize



vocabulary

token	index
剧情	0
演员	1
,	2
推荐	3
演技	4
在线	5
台词	6
紧凑	7
!	8
老套	9
强烈	10
...	...

■ Tokenization

3. Map **tokens** to **IDs** for model input

text
剧情紧凑，演员演技在线，强烈推荐!
剧情老套，台词尴尬，毫无新意。
虽然是小成本制作，但情感真挚，细节打动人心。
节奏慢得像蜗牛，两个小时都在打瞌睡。
.....



token_ids					
0	7	2	1	4	...
0	9	2	6	27	...
132	25	178	18	524	...
54	217	39	343	759	...
...					

token	index
剧情	0
演员	1
,	2
推荐	3
演技	4
在线	5
台词	6
紧凑	7
!	8
老套	9
强烈	10
...	...

Deep Learning for Text Classification



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■ Tokenization

- Example: Qwen3 vocab size: 15w+

```
from transformers import AutoTokenizer
```

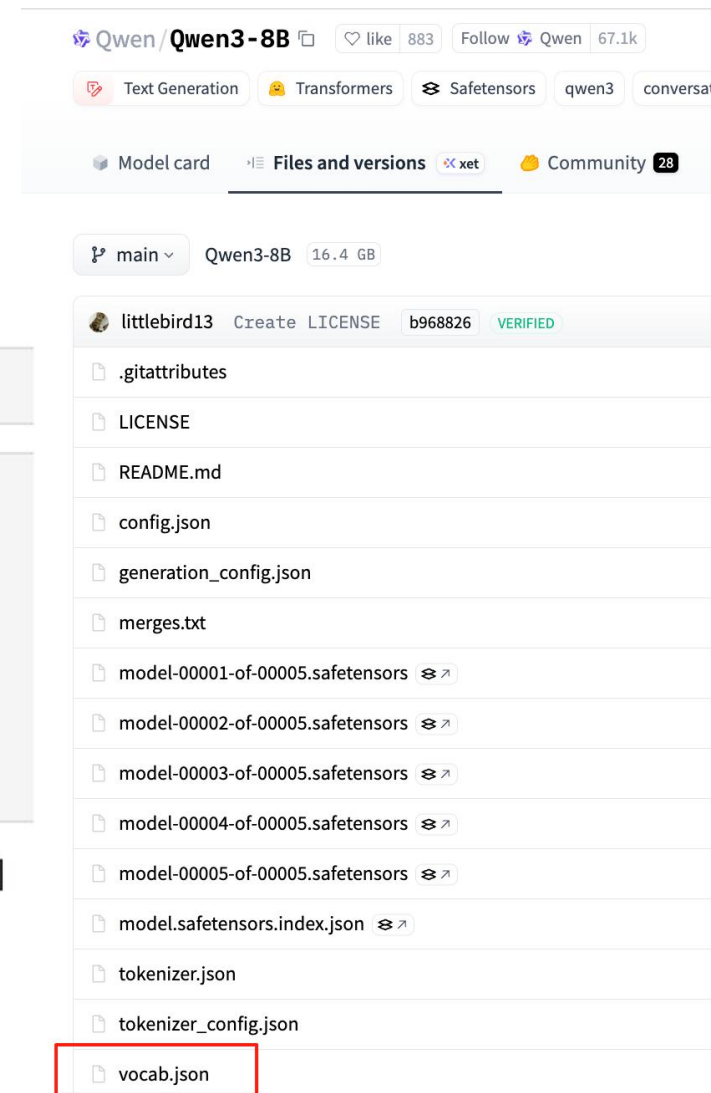
```
model_name = "Qwen/Qwen3-8B"  
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```
text = "剧情紧凑，演员演技在线，强烈推荐！"  
tokens = tokenizer(text)
```

```
tokens.input_ids
```

```
[105461, 114408, 3837, 104534, 109598, 99107, 3837, 102637, 101914, 6313]
```

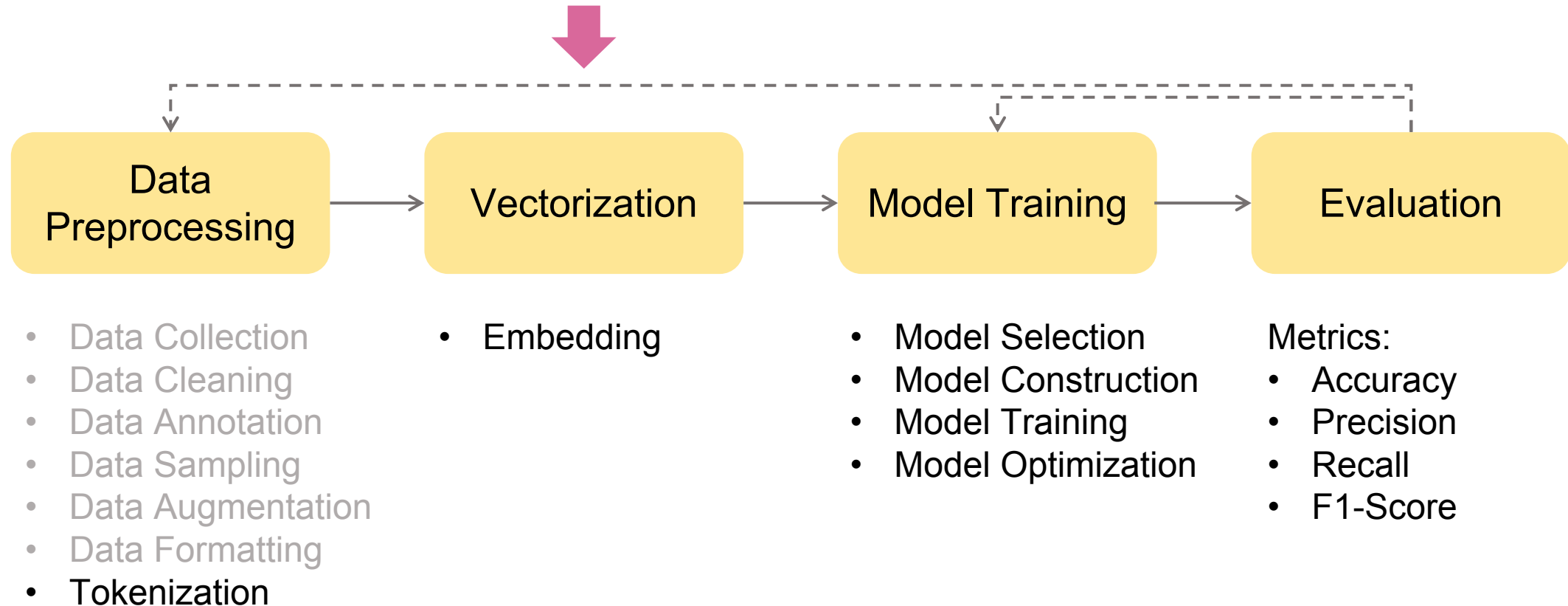
<https://huggingface.co/Qwen/Qwen3-8B/tree/main>



Deep Learning for Text Classification



■ Pipeline

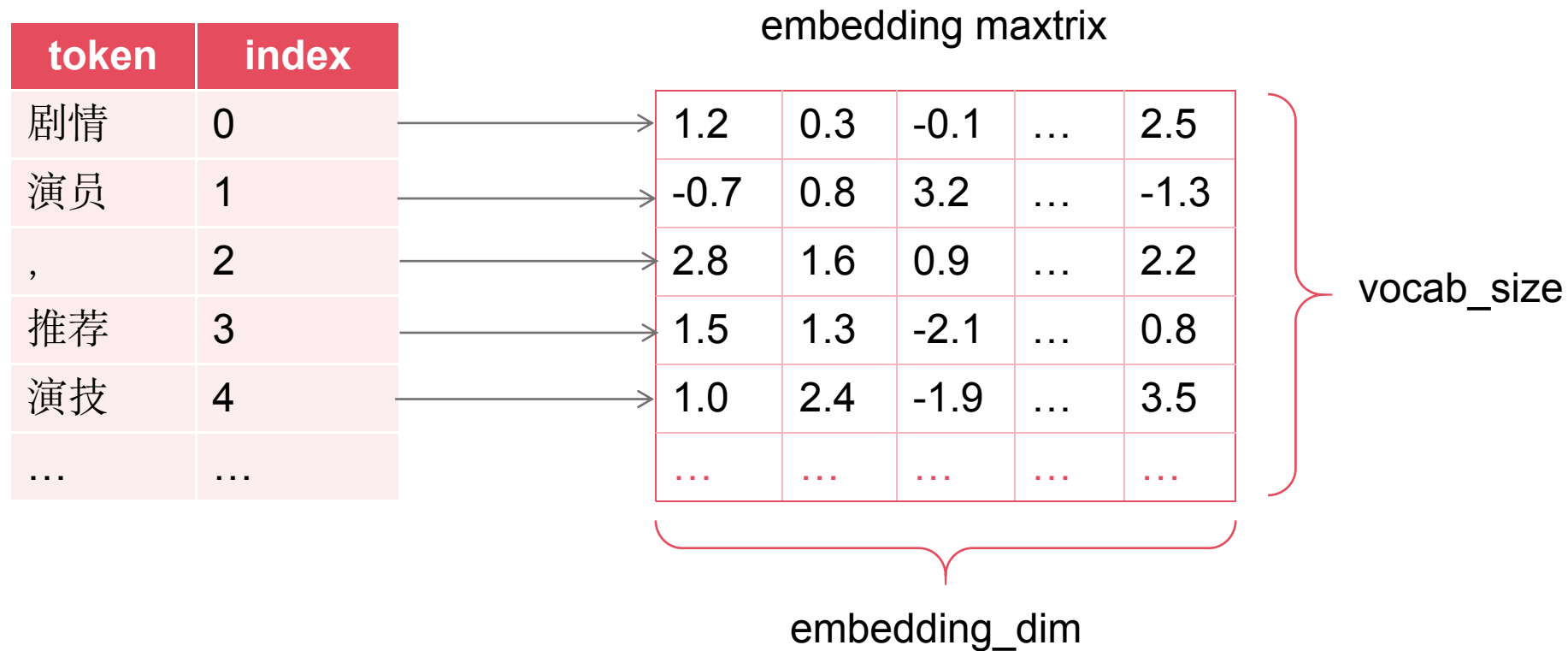


Deep Learning for Text Classification



■ Embedding

□ token ids $\rightarrow$ embeddings



■ Embedding

□ Example: Qwen3-0.6B embedding size

```
from transformers import AutoModel
```

```
model_name = "Qwen/Qwen3-0.6B"  
model = AutoModel.from_pretrained(model_name)
```

```
embedding_layer = model.get_input_embeddings()  
print("Embedding weight shape:", embedding_layer.weight.shape)
```

```
Embedding weight shape: torch.Size([151936, 1024])
```

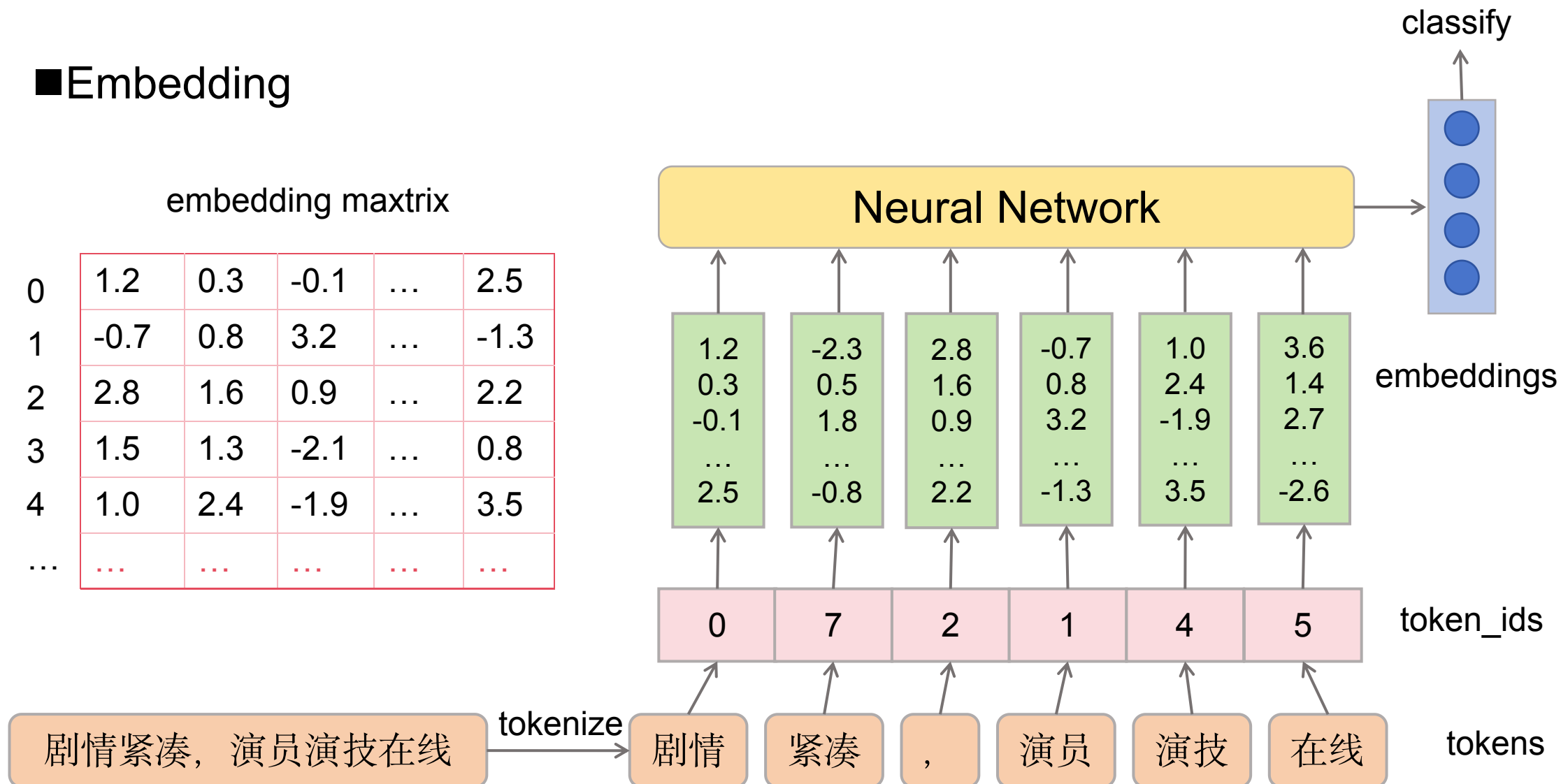
Deep Learning for Text Classification



■ Embedding

embedding maxtrix

0	1.2	0.3	-0.1	...	2.5
1	-0.7	0.8	3.2	...	-1.3
2	2.8	1.6	0.9	...	2.2
3	1.5	1.3	-2.1	...	0.8
4	1.0	2.4	-1.9	...	3.5
...	...	...	...	...	...



Deep Learning for Text Classification



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■ Embedding

□ How to get embeddings?

■ Train from scratch

■ Take pretrained word embeddings (e.g., Word2Vec, GloVe) and fix them

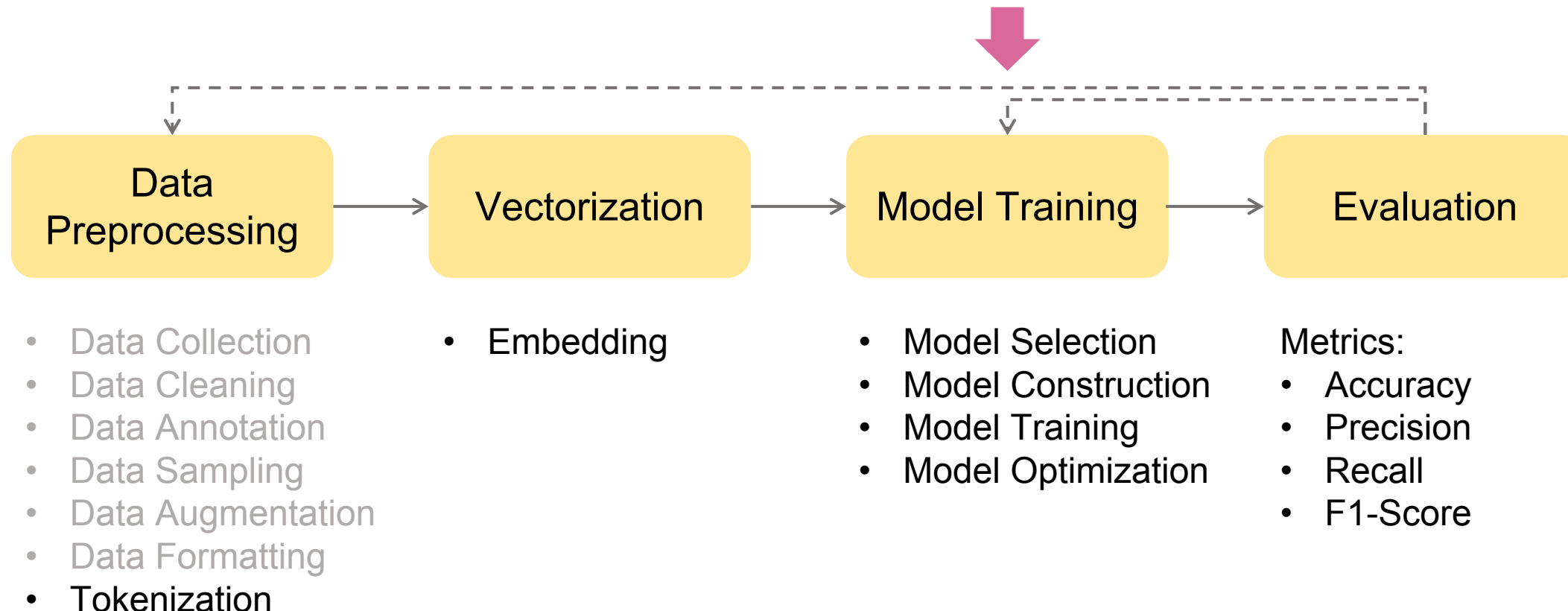
■ Initialize with pretrained word embeddings, then fine-tune

■ Initialize the whole model with a pretrained model (e.g., BERT), then fine-tune

Deep Learning for Text Classification



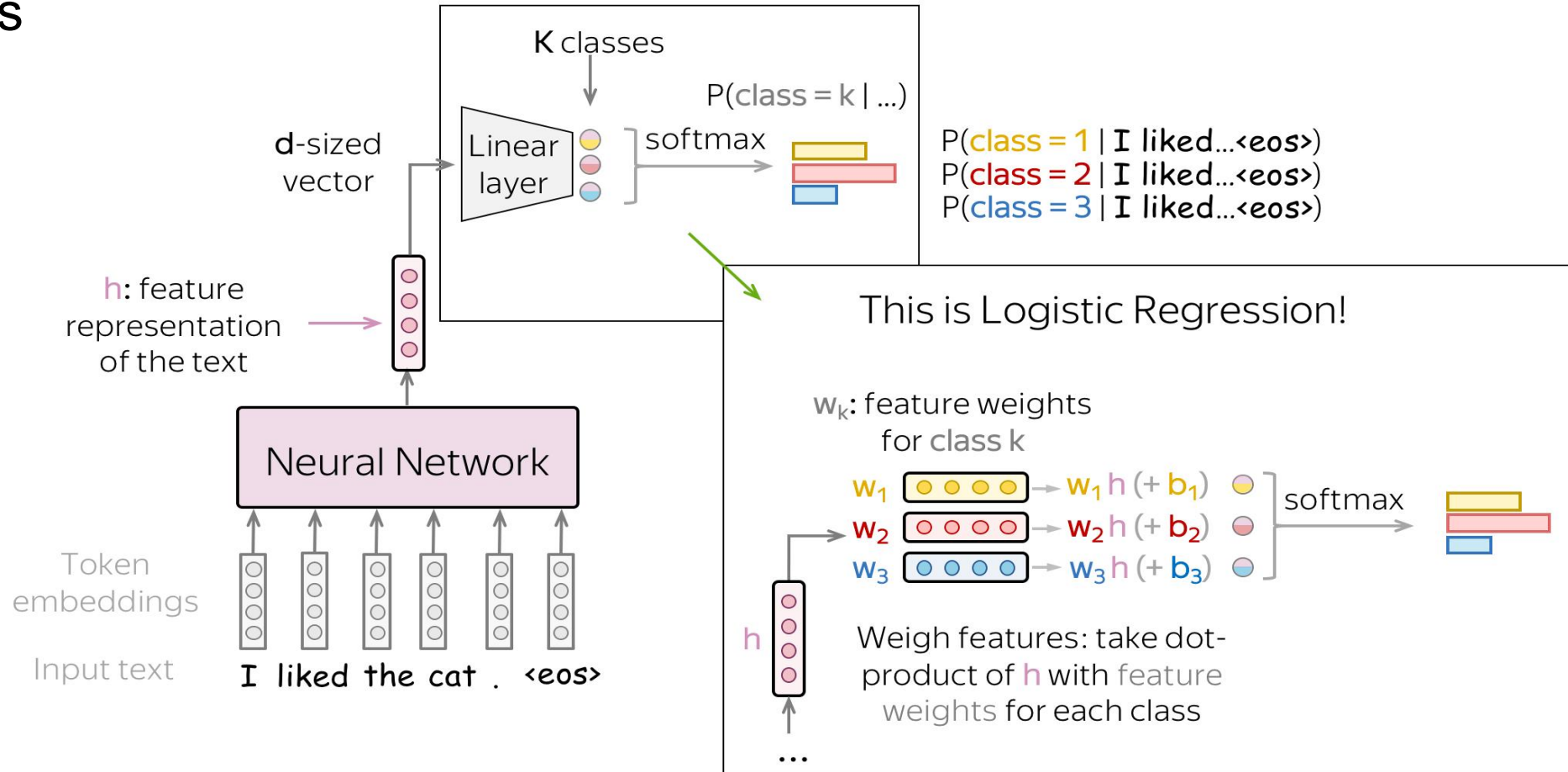
■ Pipeline



Deep Learning for Text Classification



Models

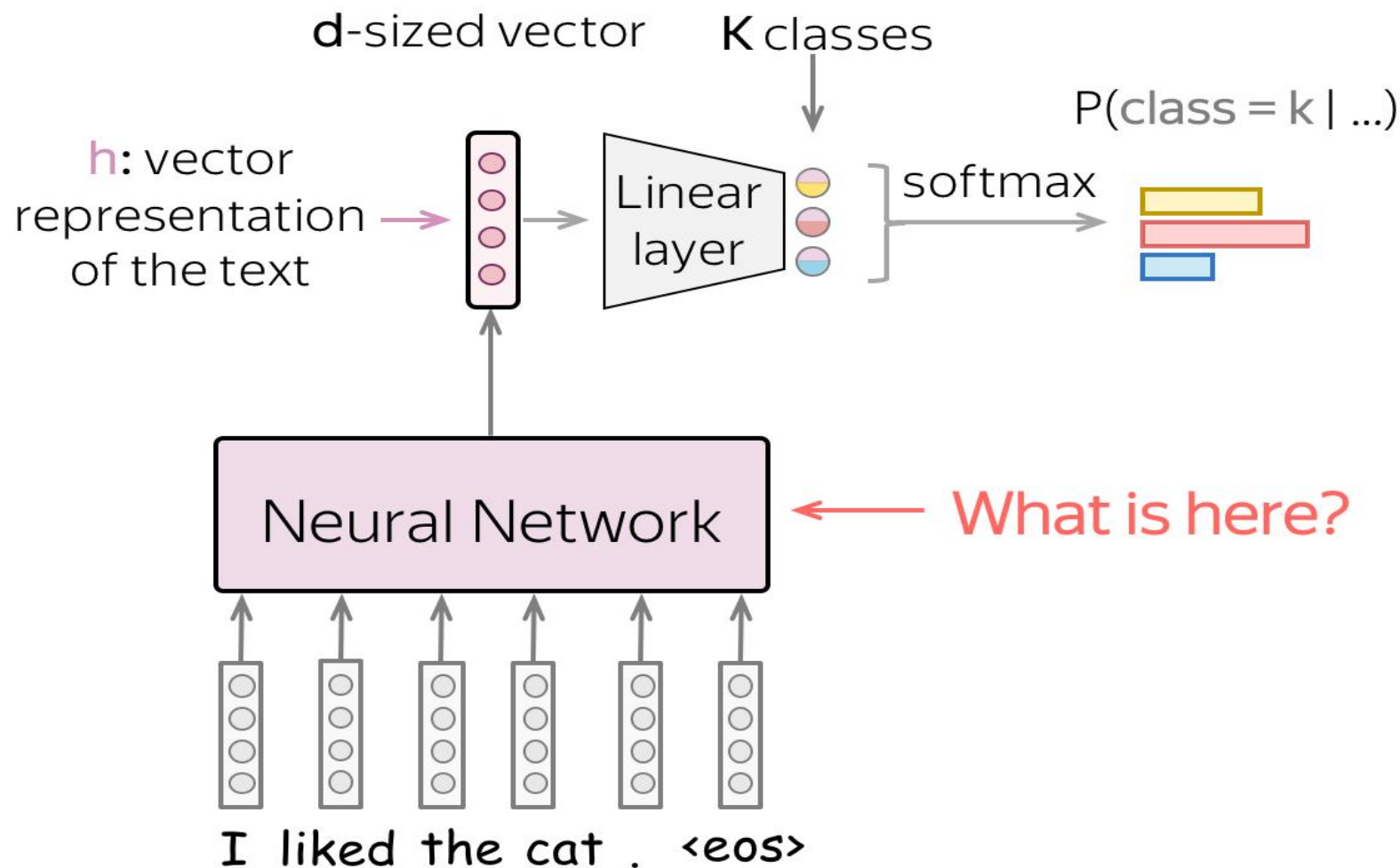


https://lena-voita.github.io/nlp_course/text_classification.html

Deep Learning for Text Classification



Models

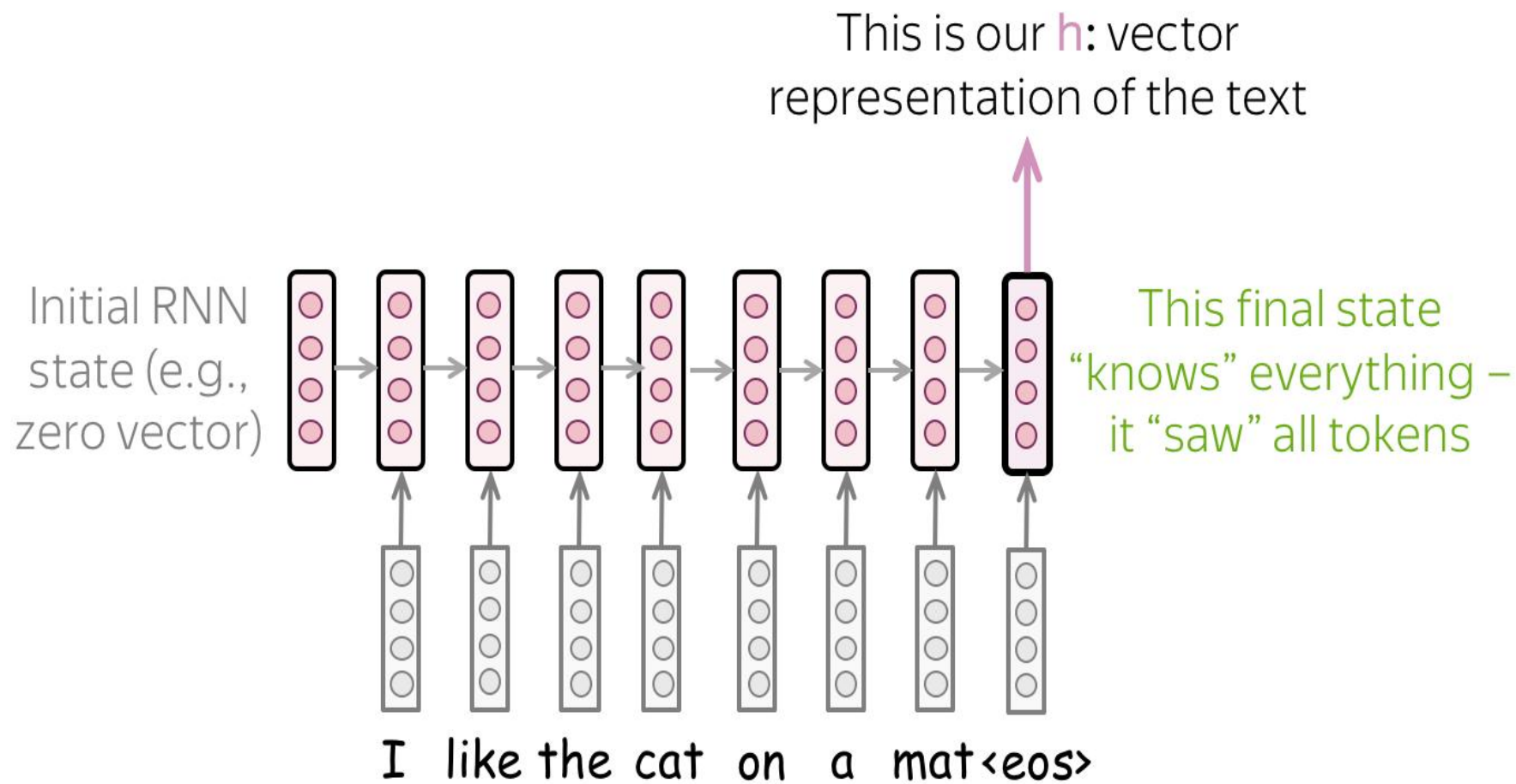


Deep Learning for Text Classification



■ Models

□ RNN

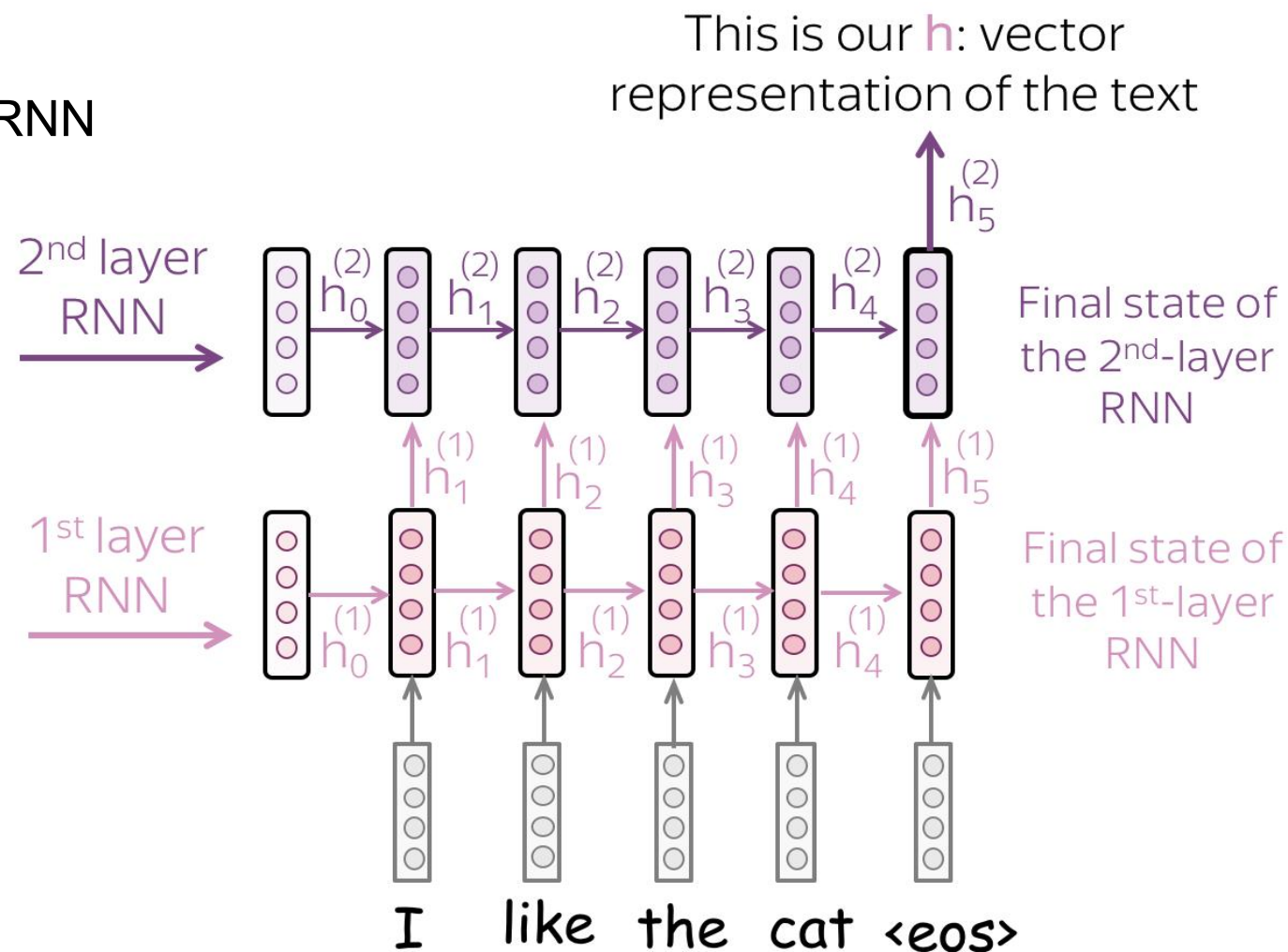


Deep Learning for Text Classification



■ Models

□ Multi-Layer RNN

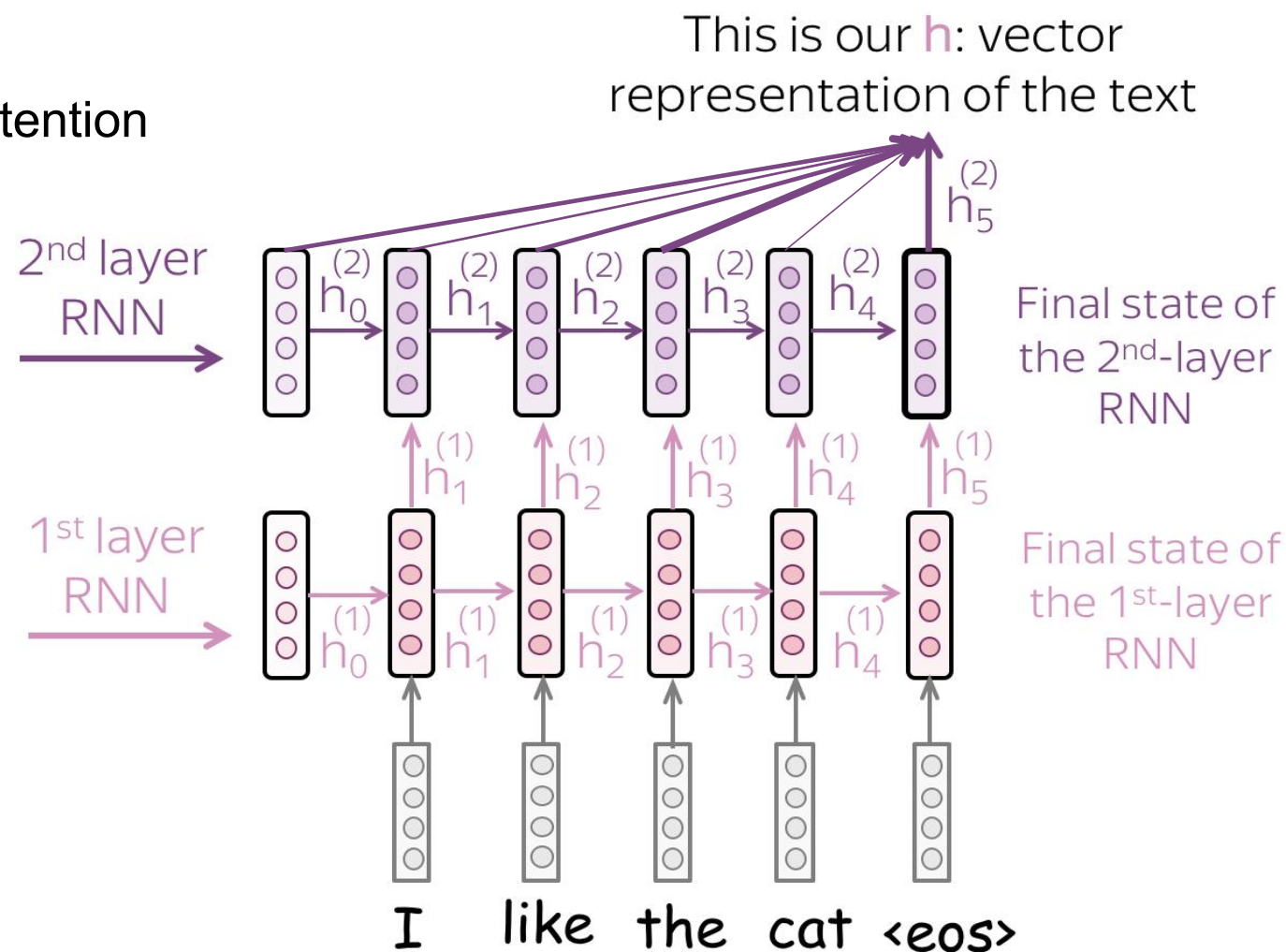


Deep Learning for Text Classification



■ Models

□ RNN with Attention



Deep Learning for Text Classification



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■ Models

□ CNN

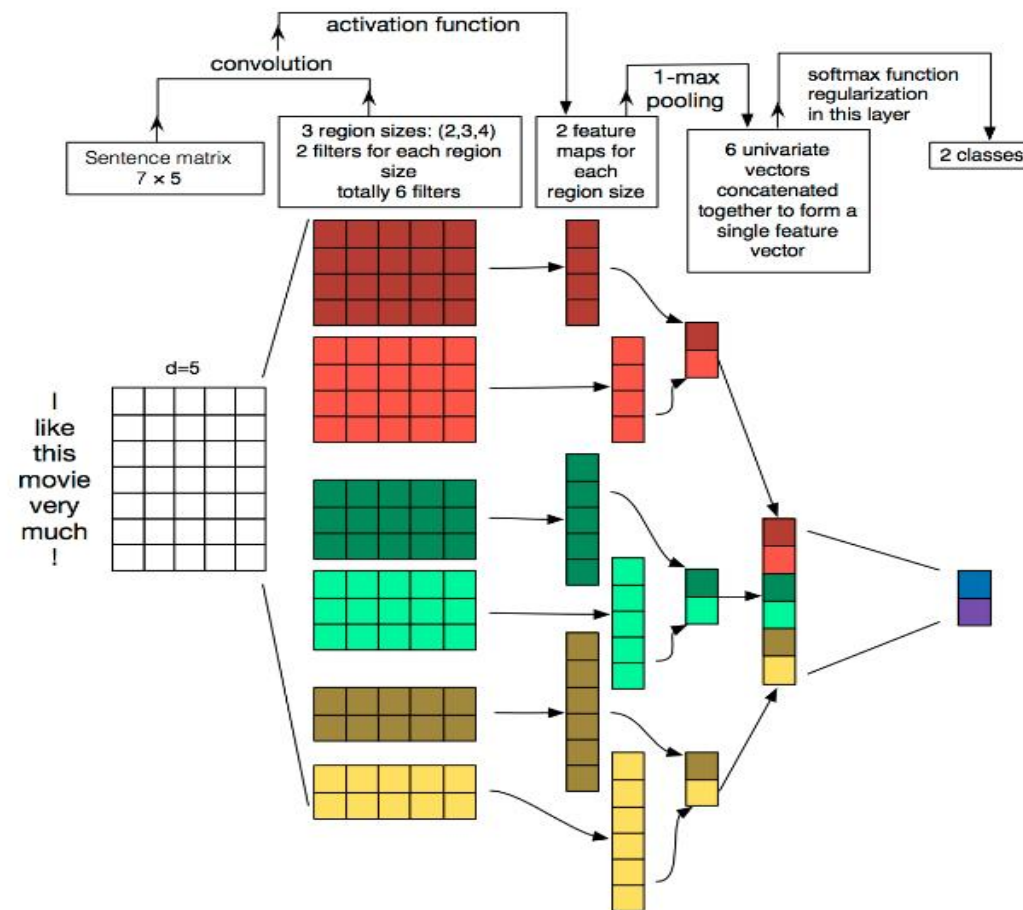


image source: <https://www.cnblogs.com/bymo/p/9675654.html>

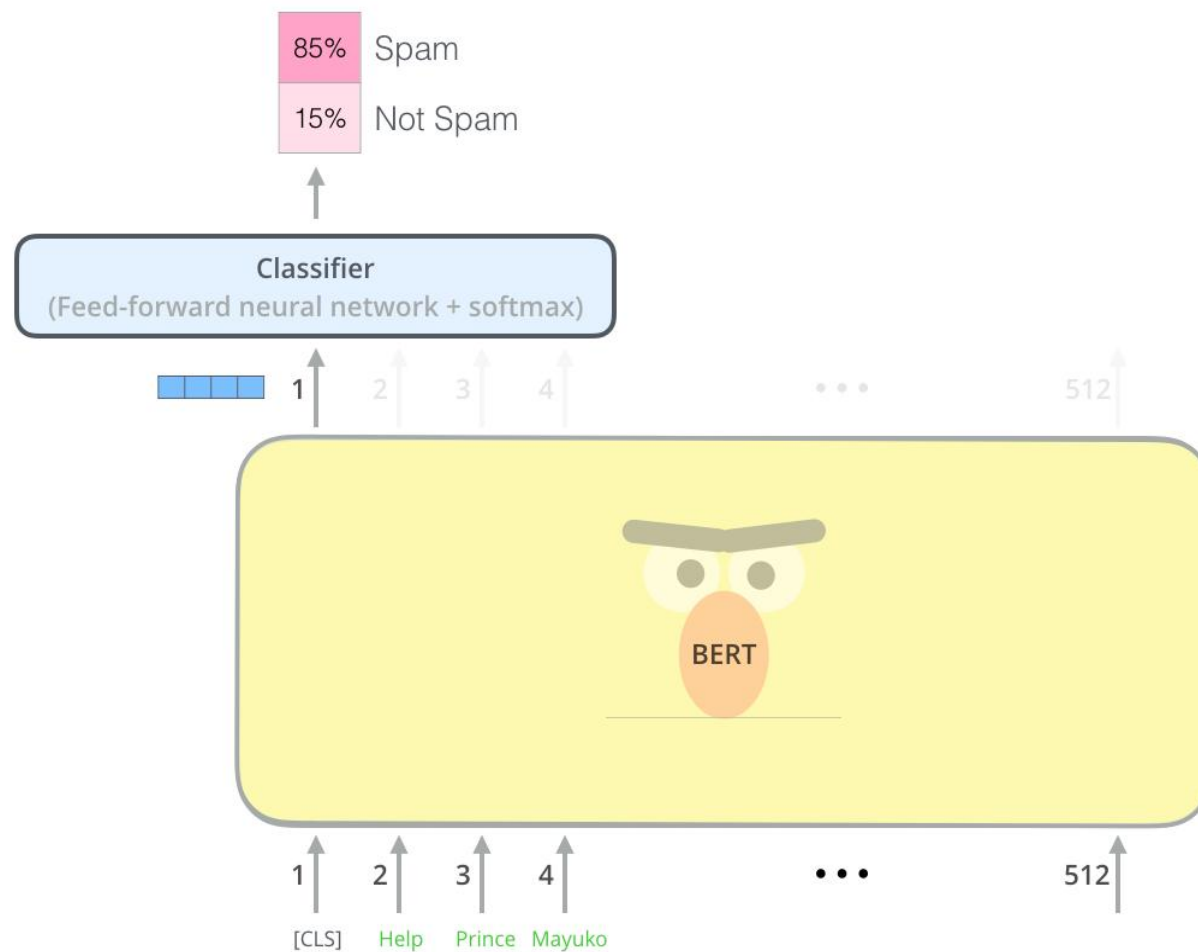
Yoon Kim. 2014. [Convolutional Neural Networks for Sentence Classification.](#)

Deep Learning for Text Classification



■ Models

□ Bert



<https://jalammar.github.io/illustrated-bert/>

■ Cross-entropy Loss

Training example: **I liked the cat on the mat** <eos>

Label: **k**
↑
target

Model prediction:

$P(\text{class} = i | \text{I liked...<eos>})$



Target:

p^*



Cross-entropy loss:

$$-\sum_{i=1}^K p_i^* \cdot \log P(y = i|x) \rightarrow \min \quad (p_k^* = 1, p_i^* = 0, i \neq k)$$

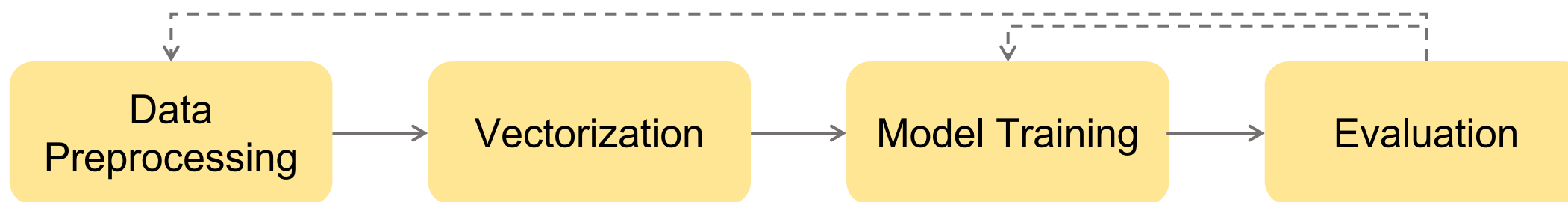
For one-hot targets, this is equivalent to

$$-\log P(y = k|x) \rightarrow \min$$

Deep Learning for Text Classification



■ Pipeline



- Data Collection
- Data Cleaning
- Data Annotation
- Data Sampling
- Data Augmentation
- Data Formatting
- Tokenization

- Embedding

- Model Selection
- Model Construction
- Model Training
- Model Optimization

- Metrics:
- Accuracy
 - Precision
 - Recall
 - F1-Score

Text Classification



■ Evaluation

$$\text{Accuracy} = \frac{\text{Number of correctly predicted samples}}{\text{Total number of samples}}$$

Is this enough?

Text Classification



■ Evaluation

Example: Imagine you build an AI system to detect whether a patient has a disease.

Ground truth: Out of 100 patients, 20 actually have the disease (positive), and 80 are healthy (negative).

AI prediction: The model says 30 patients are sick, and 70 are healthy.

	Actually Sick (Positive)	Actually Healthy (Negative)
Predicted Sick	15	15
Predicted Healthy	5	65

$$\text{precision} = 15/30 = 0.5$$

$$\text{recall} = 15/20 = 0.75$$

$$\text{accuracy} = (15 + 65)/100 = 0.8$$

Text Classification



■ Evaluation: confusion matrix

		<i>gold standard labels</i>		
		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$		accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$

Figure 4.7 A confusion matrix for visualizing how well a binary classification system performs against gold standard labels.

Daniel Jurafsky and James H. Martin. 2026. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition, Chapter 4. <https://web.stanford.edu/~jurafsky/slp3/4.pdf>

Text Classification



■ Evaluation

□ Accuracy

- Simple and intuitive
- Misleading when classes are imbalanced

□ Recall

- Measures completeness in finding real positives
- Use when false negatives are costly (e.g., missing a cancer diagnosis; failing to find relevant items in a search system)

□ Precision

- Measures reliability of positive predictions
- Use when false positives are costly (e.g., marking a legitimate email as spam)

Text Classification



■ Evaluation

□ F1-Score

- Precision and recall often trade off (improving one hurts the other), F1 gives their harmonic mean:

$$F1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

- F1-score is a balanced measure that combines precision and recall
- Best when you need both high precision and high recall

Text Classification



■ Evaluation: Multi-class Classification

		<i>gold labels</i>			
		urgent	normal	spam	
<i>system output</i>	urgent	8	10	1	precision_u = $\frac{8}{8+10+1}$
	normal	5	60	50	precision_n = $\frac{60}{5+60+50}$
	spam	3	30	200	precision_s = $\frac{200}{3+30+200}$
		recall_u = $\frac{8}{8+5+3}$	recall_n = $\frac{60}{10+60+30}$	recall_s = $\frac{200}{1+50+200}$	

Confusion matrix for a three-class classification task

Daniel Jurafsky and James H. Martin. 2026. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition, Chapter 4. <https://web.stanford.edu/~jurafsky/slp3/4.pdf>

Text Classification



■ Evaluation: Multi-class Classification

□ In order to derive a single metric that tells us how well the system is doing, we can combine these values in two ways: **Macro Average** and **Micro Average**.

$$\square \text{Macro-Precision} = \frac{1}{K} \sum_i^K \text{Precision}_i$$

$$\square \text{Macro-Recall} = \frac{1}{K} \sum_i^K \text{Recall}_i$$

$$\square \text{Micro-Precision} = \frac{\sum TP_i}{\sum (TP_i + FP_i)}$$

$$\square \text{Micro-Recall} = \frac{\sum TP_i}{\sum (TP_i + FN_i)}$$

□ Micro-Precision=Micro-Recall=Accuracy

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	precision_u = $\frac{8}{8+10+1}$
	normal	5	60	50	precision_n = $\frac{60}{5+60+50}$
	spam	3	30	200	precision_s = $\frac{200}{3+30+200}$
		recall_u = $\frac{8}{8+5+3}$	recall_n = $\frac{60}{10+60+30}$	recall_s = $\frac{200}{1+50+200}$	

Text Classification



■Evaluation: Multi-class Classification

Separate confusion matrices for the 3 classes from the previous figure

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

Pooled

	true yes	true no
system yes	268	99
system no	99	635

$$\text{microaverage precision} = \frac{268}{268+99} = .73$$

$$\text{macroaverage precision} = \frac{.42+.52+.86}{3} = .60$$

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Deep Learning for Text Classification



■ Pipeline

